

Question	Answer	Mark	Mark scheme	Additional guidance
19	$24 + 5\pi$	P1 P1 P1 A1	for process to find arc CD $\frac{40}{360} \times \pi \times 24$ oe $(= \frac{8}{3} \pi)$ for process to find arc $AF - \text{arc } BE$ $\frac{100}{360} \times \pi \times 14 - \frac{40}{360} \times \pi \times 14$ oe $(= \frac{7}{3} \pi)$ (dep P1) for “$\frac{8}{3} \pi$” + “$\frac{7}{3} \pi$” + at least 2 from OA, OF, BC, ED OR for 5π for $24 + 5\pi$ from correct calculations	Accept process to find $AB + EF$ (assumes that logo is symmetrical)